Hand Calculation: Ideal Linear Resistor

Given

Quantity	Symbol	Value
Resistance	`R`	`1 kΩ`
Evaluation voltage	`V`	`5 V`
LTspice measured current	`I_SPICE`	`4.99999988824 mA`

Analytical current

For an ideal resistor, Ohm's law gives

$$I = \frac{V}{R}$$

Therefore,

$$I_{analytical} = \frac{5\,\mathrm{V}}{1000\,\Omega} = 0.005\,\mathrm{A} = 5.000000\,\mathrm{mA}$$

Conductance

$$G = \frac{1}{R} = \frac{1}{1000\,\Omega} = 1\,\mathrm{mS}$$

The slope of the I–V curve is therefore `1 mS`.

Power at 5 V

Using the passive-resistor relationship,

$$P = VI = \frac{V^2}{R}$$

$$P_{5V} = \frac{(5\,\mathrm{V})^2}{1000\,\Omega} = 0.025\,\mathrm{W} = 25\,\mathrm{mW}$$

Analytical versus SPICE comparison

The absolute current error is

$$\left|I_{SPICE}-I_{analytical}\right| = \left|4.99999988824-5.0000000000\right|\,\mathrm{mA} = 0.00000011176\,\mathrm{mA}$$

or approximately `0.11176 nA`.

The relative error is

[\epsilonr = \frac{\left|I{SPICE}-I{analytical}\right|} {I{analytical}}\times 100 \approx 2.24\times 10^{-6}\%.]

This is numerical solver and display precision, not a physical modelling discrepancy. The ideal resistor model is linear and has no tolerance or temperature coefficient in this experiment.

Conclusion

At `V = 5 V`, the analytical current is `5 mA` and LTspice reports `4.99999988824 mA`. The result verifies Ohm’s law and the expected `1 mS` I–V slope within numerical precision. The corresponding ideal-resistor power is `25 mW`.